

SUBMISSION – 13 µL IN A 1.5 ML TUBE

3 G00101 sequencing oligo (2.66 µM)

10 miniprep DNA

13 µL total

Our sequencing oligos are standardised to **2.66 µM**, so **3 µL is exactly one reaction's worth**. The other 10 µL is DNA and water.

TUBES AND LABELS

– **1.5 mL tubes. Not PCR tubes.**

– Label the **top** of the tube.

– The label is **the exact name you put on the submission form** – not an abbreviation, not a name that only makes sense to you. The facility matches tube to form by that string.

HOW MUCH DNA – BY COPY NUMBER

– **Medium copy** – pBR322, pAC, and **pP6: 10 µL** miniprep.

– **High copy** – pUC plasmids: about **4 µL** miniprep, made up to 10 µL with water.

– **Low copy** – pSC101, BACs: **PCR the region first** and sequence the PCR product.

USE THE DGTP PROTOCOL

Not the standard chemistry. Standard chemistry **dies inside a hairpin** – a terminator, or any strong secondary structure, stops the read dead. This cost Tlib2 an entire sequencing run.

Say **dGTP** on the submission form. Results come back in **1–2 days**.

WHAT YOU GET

One primer, so extension is **linear**, not exponential. The read starts **20–50 bp downstream** of the primer and gives **400–1000 bp** of usable sequence. Pick a primer **upstream** of what you want to see – for pP6, **G00101**.

SEQUENCE EVERYTHING YOU PICKED

Including the boring-looking ones. The clones in the **middle of the range** are the informative ones. Tlib2 sequenced only its brightest and left its main question unanswered for three years.

CHECK THE READ

A .txt of base calls (the *read*) and an .ab1 trace. Open both in **ApE**; ctrl-K annotates features.

1. Is it **clean**? Length with no Ns: 100 bp poor, 800 good, 1000 great.

2. Architecture **BseRI** → **promoter** → **BseRI**. Exactly **two** BseRI sites.

3. **T4 terminator** present; promoter not duplicated, reversed or truncated.

4. Align to pP6 .seq – **Tools** → **Align with another sequence...**

5. Search for the motif below. Clean read + motif present = **usable**.

```
GAGGAGTCCTGGGTTCCNNNNNTTGACANNNNNNNNNNNNNNNNNTATAA  
TNNNNNNANNNNGTTAGTATTTCTCCTC
```

RECORD

exp · clone_id · student_name · read_name · date_sequenced · canonical · usable · cassette · notes

– **canonical** = matches pP6 .seq across the good-quality region. **usable** = contains the motif. Different questions.

– Expect artifacts – duplications, deletions, recombinations, and some plain parent plasmid. Record them as artifacts and exclude them.